

Information Security

Introduction

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Module Information

- Name: Information Security
- Year: 2
- Semester: 2nd Semester
- Credits: 3 ECTS
- Lecturers
 - Course coordinator: Juan Manuel Castelo Gómez
 - Lecturing team for the english group
 - Julio César Hernández Castro

Units (Block I)

Units	Subunits
Unit 1. Public Key Cryptography	1.1. Intro
	1.2. Symmetric vs Assymmetric Cryptography
	1.3. Diffie-Hellman Key exchange.
	1.4. Inverse calculation via Euclides
	1.5. Fast Exponentiation Algorithm
Unit 2. RSA	2.1. Basics
	2.2. Key generation
	2.3. Encryption
	2.4. Block encryption
Unit 3. Attacks on RSA	3.1. Factorization attacks
	3.2. Side-Channel Attacks
	3.3. Other attacks

Units (Block II)

Unit	Subunit
Unit 4. Elliptic Curves	5.1. Continuous and discrete elliptic curves
	5.2. Elliptic curve cryptography
	5.3. Diffie-Hellman key exchange with ECs
Unit 5. Hash functions	5.1. Hash function properties
	5.2. MD5, SHA-1 and SHA-2 functions
	5.3. Introduction to SHA-3
	5.4. Birthday paradox

Units (Block III)

Unit	Subunit
Unit 6. Digital Signatures	6.1. Digital Signatures in RSA
	6.2. DSA
Unit 7. Authentication and Digital Certificates	7.1. Authentication mechanisms
	7.2. Introduction to digital certificates
	7.3. Certification Authorities
	7.4. X.509 standard

Units (Block IV)

Unit	Subunit
Unit 8. Risk Management	8.1. Risk
	8.2. Risk Management
	8.3. Risk Treatment and Analysis
Unit 9. Information Security Management System	9.1. Introduction
	9.2. Information Security Management System
	9.3. ISMS Steps

Assessment- Regular

- **3 exams for first three blocks (units 1-7)**
 - In person: 30 minutes after the practical lecture
 - Theory and/or practical exercises
 - No minimum mark
- **Group work (units 8 and 9)**
 - 10% of final mark
 - Not ammendable
- **The exams can be retaken in the ordinary assessment (not necessary if the final grade is ≥ 5)**

Assessment - Extraordinary

- Only final exam available (exams for all blocks).
- **Group work cannot be recovered (10% of the mark)**

Regular Assessment

Sem.	Descripción	Modalidad	Tipo	Duración	Peso en la nota	Nota mínima	Competencias evaluadas
6	Evaluación Progresiva del Bloque I del temario.	EX: Técnica del tipo Examen Escrito	Presencial	00:30	30%	0 / 10	CC1
9	Evaluación Progresiva del Bloque II del temario.	EX: Técnica del tipo Examen Escrito	Presencial	00:30	30%	0 / 10	CC1
12	Evaluación Progresiva del Bloque III del temario.	EX: Técnica del tipo Examen Escrito	Presencial	00:30	30%	0 / 10	CC1
17	Actividades de la competencia Transversal	TG: Técnica del tipo Trabajo en Grupo	No Presencial	00:00	10%	0 / 10	CT8

Final only assesment

Sem	Descripción	Modalidad	Tipo	Duración	Peso en la nota	Nota mínima	Competencias evaluadas
17	Actividades de la competencia Transversal	TG: Técnica del tipo Trabajo en Grupo	No Presencial	00:00	10%	0 / 10	CT8
17	Examen de Recuperación del Bloque I del temario.	EX: Técnica del tipo Examen Escrito	Presencial	00:30	30%	0 / 10	CC1
17	Examen de Recuperación del Bloque II del temario.	EX: Técnica del tipo Examen Escrito	Presencial	00:30	30%	0 / 10	CC1
17	Examen de Recuperación del Bloque III del temario.	EX: Técnica del tipo Examen Escrito	Presencial	00:30	30%	0 / 10	CC1

Extra Assessment

Descripción	Modalidad	Tipo	Duración	Peso en la nota	Nota mínima	Competencias evaluadas
Examen "Sólo prueba final" de todo el contenido de la asignatura	EX: Técnica del tipo Examen Escrito	Presencial	02:30	90%	0 / 10	CC1



10% obtenido en la semana 13

Teaching



Theory

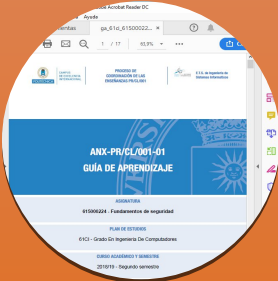
- Lectures
- Resources
 - Slides
 - Moodle extras



Practical exercises

- Solved problems and exercises
- Resources
 - Solved problems
 - Software

Resources I



Module Guide



Moodle
([UPM-Estudios Oficiales](#))



Bibliography



Resources II

Forums and Office hours



✓ Fechas de Exámenes/Exams Dates ✎

Los exámenes de la asignatura se realizarán en las siguientes fechas: / The course's exams will be held in the following dates:

Bloque I (Temas 1-3)/Block I (Units 1-3)

- Grupo IWSIT21: Lunes 9 de marzo de 17-19. Aula por determinar
- Grupo IWSIM21: Martes 10 de marzo de 9-11. Aula por determinar / IWSIM21 Group: Tuesday March 10th from 9-11. Room to be confirmed

Learning guide summary:

- The evaluation is based in **three block exams** (30% each) and a **group work** (10%).
- **There is no minimum grade in neither the exams nor the group work.** To pass the course, a final grade equal to or above 5 is necessary.
- **Failed block exams can be retaken in the ordinary assessment.** It is not necessary to do so if the final grade is equal to or above 5.
- The **extraordinary assessment consists of a single exam for all blocks.**
- **The group work cannot be retaken.**

✓ Exámenes de Cursos Anteriores / Exams from Previous Years ✎

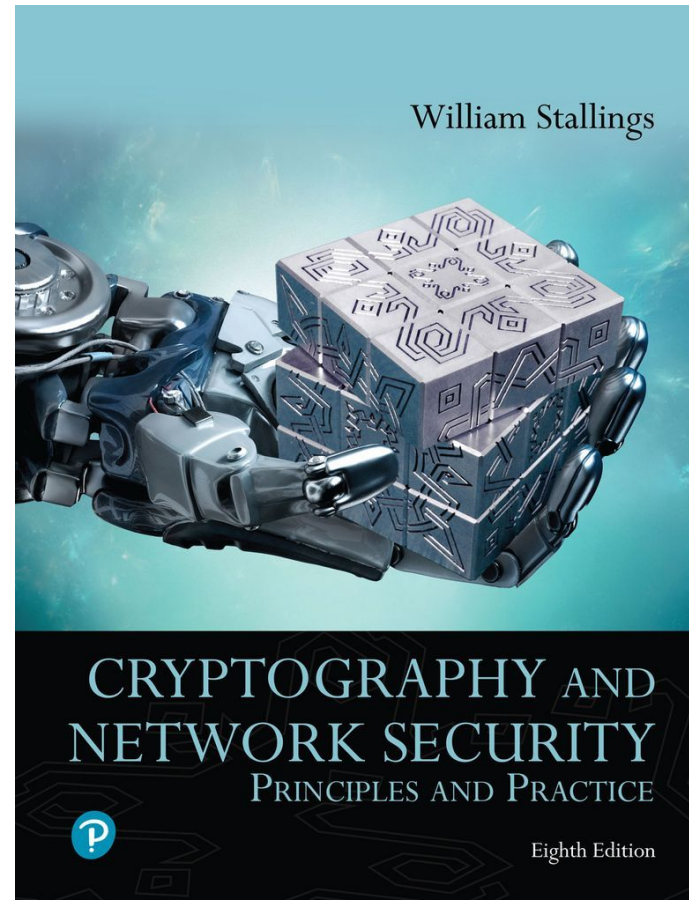


 Soluciones Exámenes Parciales 2025 / Exams from 2025 ✎

 Soluciones Exámenes 2024 / Exams from 2024 ✎

 Soluciones Exámenes 2023 / Exams from 2023 ✎

Bibliography



Questions?

